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Mr. John Steve C. Landiza
Mr. Scott Alao
Teaching Interns

February 25, 2026
Date of Teaching

Mr. Sunny R. Fernandez
Teaching Internship Mentor

Grade 5/ WTh 7:30 AM - 10:00 AM
Class and Schedule

February 23, 2026
Date of Submission

February , 2026
Date of Approval

LESSON PLAN in Math 5
(1st Revision)

Content Standard	*The learners should have knowledge and understanding of* ● GMDAS rules when performing three or more operations with fractions and decimals.
Performance Standard	*By the end of the quarter, the learners are able to* ● apply the GMDAS rules with operations with fractions and decimals.
Learning Competencies	*The learners:* a. solve multi-step problems involving division of decimals that may or may not also involve the other operations with decimals, including problems involving money. b. perform three or more different operations with fractions and decimals by applying the GMDAS rules.

I. Intended Learning Outcomes (Objectives)

At the end of the lesson, the Grade 5 learners, with at least 80% accuracy, will be able to:

Knowledge	identify the correct order of operations when presented with problems involving fractions;
Skills	solve multi-step problems involving fractions by using GMDAS;
Attitudes/Values	value the importance of GMDAS in avoiding errors when handling real-world transactions.

Certification Date: 24 January 2024
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For verification of the certificate please Access www.gcl-intl.com (Certification Check and type the registration number)
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Times Higher Education Impact Rankings

II. Learning Content	
A. Topic (s)	**Lesson 4.2 GMDAS on Fractions**
B. Key Concepts	1. In solving expressions with three or more operations involving decimals, follow the correct order: *Grouping symbols, (Multiplication and Division) whichever appears from left to right, (Addition, and Subtraction) whichever appears from left to right. (GMDAS).* 2. To solve multi-step problems involving fractions, follow these steps: *a. What is given?* Identify the facts, numbers, and important information stated in the problem. *b. What is asked?* Determine what the problem requires you to find. *c. What are the operations to be used?* Identify the correct operations involved (division, multiplication, addition, subtraction) and apply the GMDAS rule. *d. What is the number sentence?* Translate the word problem into a mathematical expression or equation. *e. Solve.* Perform the operations step-by-step following GMDAS and proper decimal procedures. *f. What is the complete answer?* State the final answer clearly with the correct unit (if any) and in a complete sentence.
C. References *(Print, electronic, etc.)*	- Gureng, P. T., Jimenez, E. P., De Gracia, M. M. B., Ditablan, J. T., Gromio, E. G., Perez, R. M., & Tuliao, A. L. *(2025). Realistic Math: Basic, Beyond, Breakthroughs* (Second Edition), Phoenix Publishing House, Inc., pages 256–265

D. Instructional Materials	• Laptop • Canva Presentation • Google Docs
E. Integration *(Values, SDG/ESD, GAD, etc.)*	**Values:** Accuracy and Financial Responsibility **GAD:** Shared Responsibility in Financial Management **SDG 4:** Quality Education
F. Skills *(21st century skills)*	• Critical Thinking and Problem Solving • Communication Skills • Collaboration

III. Teaching and Learning Process (Experiences)

Model: 5E Instructional Model
Total time allotment: 90 minutes

PRELIMINARIES (5 minutes)

The teacher begins the lesson with a prayer, followed by attendance checking. The teacher will introduce themselves as one of the new guides for the students in the last quarter of the school year. The teacher will provide a brief recap of lesson 4.1: GMDAS on decimals, the proper order of operations, and the steps in solving problems.

A. ENGAGE (15 minutes)

The teacher will activate their prior knowledge about the topic by presenting a sample number problem. Afterwards, the teacher will give 3 minutes or ample time for the students to solve.

$$19 + 35 \div 5 \times (2 + 3)$$

With this number problem, the teacher will recap GMDAS.

This is to be followed by a recap of operations on fractions.

B. EXPLORE (15 minutes)

The teacher will focus on the different use cases of fractions vs decimals. The teacher will first show a problem, and ask the students to answer using decimals:

A baker has 5 identical cakes. He divides them equally among 3 display tables. The staff at one table immediately sells ½ of their allocated cake. What exact amount of cake remains on that table?

Decimals:

Given: 5 cakes, 3 tables, 0.5 of allocated cakes sold
Asked: **Exact** amount of cake on that table
Operations: Division, Multiplication
Number Sentence: (5÷3) x 0.5 **=** Exact amount of cake
Solution: 1.66666666666… x 0.5 = 0.8333333333333333333333333…
Answer: There is exactly 0.8333333333333333333333333… cakes left on their table.

Now the teacher will do the same problem but in fraction form:

Given: 5 cakes, 3 tables, ½ of allocated cakes sold
Asked: **Exact** amount of cake on that table
Operations: Division, Multiplication
Number Sentence: (5÷3) x ½ **=** Exact amount of cake
Solution: $\frac{5}{3}x\frac{1}{2} = \frac{5}{6}$

Answer: The exact amount of cake left on that particular table is ⅚ cakes.

After the short problem solving, the teacher will be asking the students processing questions:

1. Examine the decimal calculation for 5÷3. Why is it mathematically impossible to write the exact final decimal?
2. Analyze the fraction calculation. Did the fraction method force you to round, estimate, or cut off any numbers?
3. Compare the two final answers. What happens to the accuracy of our final answer if we use decimals for numbers that do not terminate?
4. Based on this evidence, state the rule: When are you strictly required to use fractions instead of decimals to solve a mathematical problem?

C. EXPLAIN (20 minutes)

The teacher will formally define the GMDAS rule for fractions and present the mandatory six-step problem-solving framework.

The teacher will instruct the learners that to solve multi-step problems involving fractions, they must follow these steps:

a. What is given? Identify the facts, numbers, and important information.

b. What is asked? Determine what the problem requires you to find.

c. What are the operations to be used? Identify the correct operations and apply the GMDAS rule.

d. What is the number sentence? Translate the word problem into a mathematical expression.

e. Solve. **Before solving, all mixed numbers must be turned into improper fractions first.** Perform the operations step-by-step following GMDAS.

f. What is the complete answer? State the final answer clearly with the correct unit in a complete sentence.

The teacher will display a word problem and instruct the learners to execute the steps:

A baker has 5 ½ cups of flour. He uses $1\frac{1}{2} + 2\frac{1}{4}$ cups for a cake, and divides the remaining flour equally into 2 smaller bowls. How much flour is in each bowl?

Given: 5 ½ cups of flour, $1\frac{1}{2} + 2\frac{1}{4}$ cups for a cake, 2 smaller bowls
Asked: Flour in each bowl (N)
Operations: Addition, Subtraction, Division.
Number Sentence: (5 ½ - ($1\frac{1}{2} + 2\frac{1}{4}$) = N

Solution: Turn everything into proper fractions first!

5 ½ = 11/2 | 1½ +2¼ = 5/2 + 9/4

(11/2 - (3/2 + 9/4)) ÷ 2 = N
(11/2 - (15/4)) = N
(22/4 - 15/4) = N
7/4 = N

Answer: The amount of flour left is just 7/4

The teacher will display the another word problem and instruct the learners to execute the steps:

A tailor has 4½ yards of fabric. She divides the fabric into equal sections that are ¾ of a yard long. She then uses ⅔ of these sections to make shirts. How many sections did she use?

Given: 4 ½ yards of fabric, ¾ yard sections, ⅔ of sections used.

Asked: Number of sections used to make shirts. (N)

Operations: Division, Multiplication.

Number Sentence: (4½ ÷ ¾) * ⅔ = N

Solution:

1. Convert all mixed numbers to improper fractions: 4 ½ = 9/2
2. Substitute into the expression: (9/2 ÷ ¾) * ⅔
3. Evaluate from left to right. Execute division first:

$$\frac{9}{2} \div \frac{3}{4} = \frac{9}{2} \cdot \frac{4}{3} = \frac{36}{6} = 6$$

$$6 \cdot \frac{2}{3} = 4$$

Answer: She used 4 sections of the fabric she originally had.

D. ELABORATE (20 minutes)

The teacher will transition from direct instruction by stating that true mathematical comprehension requires the ability to detect and resolve errors in logical sequences. The teacher will showcase multiple expressions and/or equations that the students will have to evaluate the correctness of.

Item 1:Determine if the answer is correct. If not, find the error/s in the solution, and write the correct solution and answer.

(1) $7 - \frac{5}{9} + \frac{1}{2} \div \frac{2}{3} \times 4 = N$

(2) $7 - \left(\frac{5}{9} + \frac{1}{2}\right) \div \left(\frac{2}{3} \times 4\right) = N$

(3) $7 - \left(\frac{19}{18}\right) \div \left(\frac{8}{3}\right) = N$

(4) $\frac{107}{18} \div \frac{8}{3} = N$

(5) $\frac{321}{144} = N$

(6) $2\frac{33}{144} = N$

(7) $2\frac{11}{48} = N$

Item 2: Determine if the equation is true. If not, find the error/s in the solution, and write the correct solution.

(1) $4\frac{2}{3} \div 5 \times \frac{6}{7} = 4\frac{2}{3} \div \left(5 \times \frac{6}{7}\right)$

(2) $\frac{14}{3} \div 5 \times \frac{6}{7} = \frac{14}{3} \div \left(5 \times \frac{6}{7}\right)$

(3) $\frac{14}{3} \div \frac{30}{7} = \frac{14}{3} \div \left(\frac{30}{7}\right)$

(4) $\frac{98}{90} = \frac{98}{90}$

Item 3: Determine if the answer is correct. If not, find the error/s in the solution, and write the correct solution and answer.

(1) $5\frac{1}{3} \div \frac{2}{7} - \frac{5}{2} + \frac{6}{7} = N$

(2) $\frac{15}{3} \div \frac{2}{7} - \frac{5}{2} + \frac{6}{7} = N$

(3) $\left(\frac{15}{3} \div \frac{2}{7}\right) - \left(\frac{5}{2} + \frac{6}{7}\right) = N$

(4) $\left(\frac{15}{3} \times \frac{2}{7}\right) - \left(\frac{5}{2} + \frac{6}{7}\right) = N$

(5) $\left(\frac{30}{21}\right) - \left(\frac{11}{9}\right) = N$

(6) $\frac{21}{12} = N$

E. Evaluate (15 minutes) - Scott

The teacher begins the evaluation phase through Google forms. The google forms will contain 10 multiple choice items, each worth 1 point, 3 word

problems, each worth 5 problems, and 2 items for error analysis.

IV. Learning Assessment

Multiple Choice (1 points each x 10 = 10)

Instructions : Read each item carefully before choosing your answer.

1. According to the GMDAS rule, how do you determine whether to multiply or divide first in an expression without parentheses?

A. Multiplication is always performed before division because multiplication appears first before division.
B. Division is always performed before multiplication.
C. You perform whichever operation appears first from left to right.
D. You perform the operation with the larger numbers first to simplify them.

2. What is the first step you must take when solving a mathematical expression that contains fractions?

A. Add the whole numbers first before doing the other operations
B. Convert mixed numbers into improper fractions
C. Multiply the denominators and the numerators of each fraction
D. Find the reciprocal of the second number

3. Which of the following represents the correct sequence of the GMDAS rule?

A. Grouping, Multiplication, Division, Addition, Subtraction
B. Grouping, Addition/Subtraction, Multiplication/Division
C. Grouping, Multiplication/Division, Addition/Subtraction
D. Addition, Multiplication, Grouping, Division

4. When solving a multi-step problem, what is the purpose of the "Number Sentence" step?

A. To translate the word problem into a mathematical expression
B. To practice writing in English so our answer will have perfect grammar
C. To count how many numbers are in the problem, and to determine the operations that will be used.
D. To write the final answer with units.

5. If a problem involves a non-terminating decimal (like 0.333...), why is it recommended to use fractions instead?

A. Because fractions are always smaller.
B. Because decimals are not used in Math 5.

C. Because fractions are easier to multiply than decimals.
D. To maintain 100% accuracy.

6. Calculate the exact value of the expression: $8 - 3\frac{1}{2} \div \frac{5}{6} \times \frac{3}{4} + 7$

A. $11\frac{17}{20}$
B. $9\frac{2}{5}$
C. $9\frac{17}{20}$
D. $14\frac{1}{5}$

7. Analyze the following solution. At which step did the mathematical violation occur?

Number Sentence: $4 \div \frac{2}{3} \times 2 = N$

Step 1: $\frac{4}{1} \div \frac{2}{3} \times \frac{2}{1} = N$

Step 2: $\frac{4}{1} \div \frac{4}{3} = N$

Step 3: $\frac{4}{1} \times \frac{3}{4} = N$

Step 4: $\frac{12}{4} = N$

Step 5: 3 = *N*

A. Step 1
B. Step 2
C. Step 3
D. Step 4

8. Evaluate both sides of the following equation: $8 \div 2 \times 4 = 8 \div (2 \times 4)$

Which statement is true regarding the equation above?

A. The equation is True because 16 = 16.
B. The equation is True because multiplication always comes before division.
C. The equation is False because 4 ≠ 1.
D. The equation is False because 16 ≠ 1.

9. A vendor buys 10.5 kilograms of sugar. She uses ⅕ of it to make syrup. How many kilograms of sugar are left?

A. 8.4 kg
B. 2.1 kg
C. 10.3 kg
D. 8.0 kg

10. Analyze the equation: 12 ÷ 4 times 3 = N. A student solves it and concludes that N = 1. What specific GMDAS rule did the student violate?

A. They divided before adding.
B. They multiplied 4 with 3 first.
C. They failed to use grouping symbols.
D. They divided 12 with 4 first.

Problem Solving

Instructions: Answer the following problems following the correct process. Identify the given, asked, operations, number sentence, solution, and answer.

1. A computer hard drive has 120 ⅕ gigabytes of free space. A user installs 3 updates that each require 5.25 gigabytes. The user then deletes an old file, freeing up 2 ¾ gigabytes. How much free space remains on the drive?

2. A vendor earns ₱450.50 from selling mangoes. She gives ½ of her earnings to her mother for household expenses. She then divides the remaining money equally among her 4 children for their school allowance. How much exact money does each child receive?

3. A carpenter buys 3 identical boards that are each 4 ⅓ feet long. He divides the total length of the wood into equal sections that are exactly ⅚ of a foot long. He then gives away 7 of these sections to an apprentice. How many sections of wood does the carpenter have left?

V. Asynchronous Task (Enrichment or Agreement)

Error Analysis

Determine the correctness of each item. Write “Correct” at the last step if the solution contains no errors, and the final answer is correct. If there is an error in the solution or the answer, **determine** the step where the **first** violation occurred. **Explain** the error, and **write** the correct solution and answer.

Example item:

(1) $\frac{7}{8} \div 4\frac{3}{5} + 5 \times \frac{2}{9} = N$

(2) $\frac{7}{8} \div \frac{23}{5} + 5 \times \frac{2}{9} = N$

(3) $\frac{161}{40} + \frac{5}{1} \times \frac{2}{9} = N$

(4) $\frac{161}{40} + \frac{10}{9} = N$

(5) $\frac{1849}{360} = N$

Error: Step 3

Explanation: Error in dividing fractions. The solution did not take the reciprocal of the divisor

Solution:

$\frac{7}{8} \div 4\frac{3}{5} + 5 \times \frac{2}{9} = N$

$\frac{7}{8} \div \frac{23}{5} + 5 \times \frac{2}{9} = N$

$\frac{7}{8} \times \frac{5}{23} + 5 \times \frac{2}{9} = N$

$\frac{35}{184} + \frac{5}{1} \times \frac{2}{9} = N$

$\frac{35}{184} + \frac{10}{9} = N$

$\frac{2155}{1656} = N$

Item 1.

(1) $\frac{6}{7} \times \frac{7}{6} \div 6 + 7 = \frac{6}{7} \div \frac{6}{7} \times \frac{1}{6} + 7$

(2) $\frac{6}{7} \times \frac{7}{6} \div \frac{6}{1} + 7 = \frac{6}{7} \div \frac{6}{7} \times \frac{1}{6} + 7$

(3) $\frac{6}{7} \times \frac{7}{6} \times \frac{1}{6} + 7 = \frac{6}{7} \times \frac{7}{6} \times \frac{1}{6} + 7$

(4) $\frac{42}{42} \times \frac{1}{6} + 7 = \frac{42}{42} \times \frac{1}{6} + 7$

(5) $1 \times \frac{1}{6} + 7 = 1 \times \frac{1}{6} + 7$

(6) $\frac{1}{6} + 7 = \frac{1}{6} + 7$

(7) $\frac{49}{42} = \frac{49}{42}$

Item 2:

(1) $3\frac{1}{2} + 1\frac{1}{4} \times \frac{2}{5} = N$

(2) $\frac{6}{2} + \frac{5}{4} \times \frac{2}{5} = N$

(3) $\frac{6}{2} + \frac{10}{20} = N$

(4) $\frac{70}{20} = N$

(5) $\frac{7}{2} = N$

Item 3:

(1) $3\frac{1}{3} - 2\frac{2}{3} \times 1\frac{3}{3} \div 3 = N$

(2) $\frac{10}{3} - \frac{8}{3} \times \frac{6}{3} \div \frac{3}{1} = N$

(3) $\frac{10}{3} - \frac{8}{3} \times \frac{6}{3} \times \frac{1}{3} = N$

(4) $\frac{2}{3} \times \frac{6}{3} \times \frac{1}{3} = N$

(5) $\frac{12}{27} = N$

(6) $\frac{4}{9} = N$

Challenge:

(1) $4 \times 3\frac{2}{3} - 2 = N$

(2) $(4 \times 3) + (4 \times \frac{2}{3}) - 2 = N$

(3) $12 + \frac{8}{3} - 2 = N$

(4) $10 + \frac{8}{3} = N$

(5) $\frac{38}{3} = N$